The University Of Azad Jammu & Kashmir, Muzaffarabad



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Course: **Software Construction**

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Lab#4

Q1. Online Doctor Appointment Management System

The system allows patients to create accounts and request appointments with doctors. Doctors review requests and accept them when a suitable time slot is available. If the doctor is busy, the request is rejected. When a request is accepted, the system sends a confirmation notification to the patient.

Classes and brief explanations

1. Patient: Stores patient account details such as patient ID, name, email and password. A patient can create an account and request an appointment.

2. Doctor: Stores the doctor's details, including doctor ID, name and specialization. A doctor can view appointment requests and accept or reject them.

3. Appointment: Represents a patient's appointment request. It stores the appointment ID, requested date/time and status (Pending, Accepted or Rejected).

4. Schedule / TimeSlot: Represents the doctor's available time slots. It helps the system check whether the requested time is free.

5. Hospital / Clinic: Stores the hospital or clinic details, such as its ID, name and address, where the appointment will take place.

6. Notification: Stores and sends a message to the patient. A confirmation notification is sent when the doctor accepts the appointment.

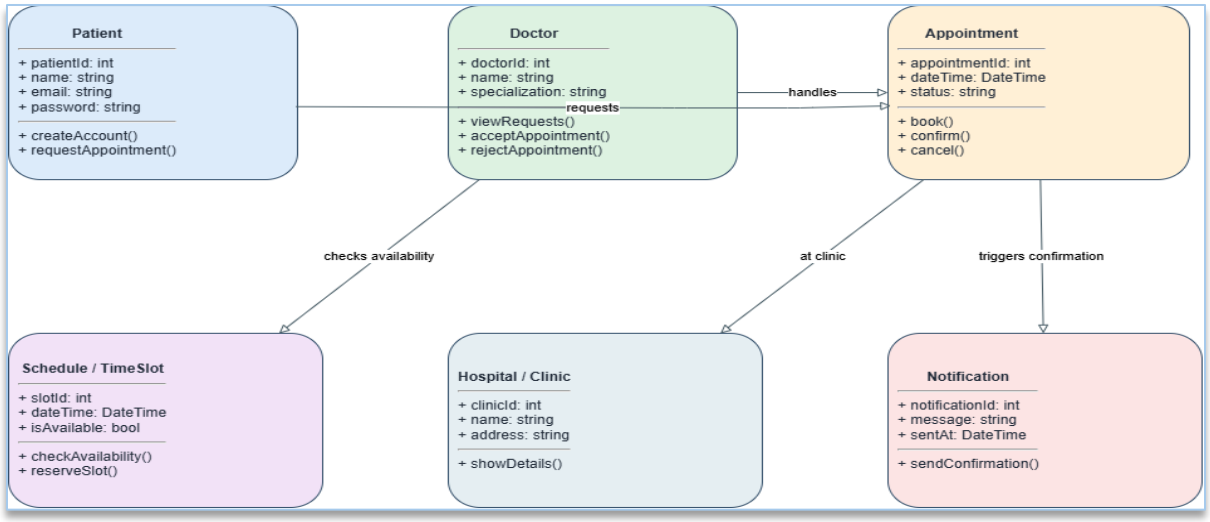


Fig: UML diagram for doctor appointment system

Q2. Animal, Cat and Dog — Inheritance

A superclass is a parent class whose common features can be inherited by other classes. Here, Animal is the superclass. Cat and Dog are subclasses. Both subclasses inherit eat() and sleep() from Animal. Cat has its own meow() function, while Dog has its own bark() function.

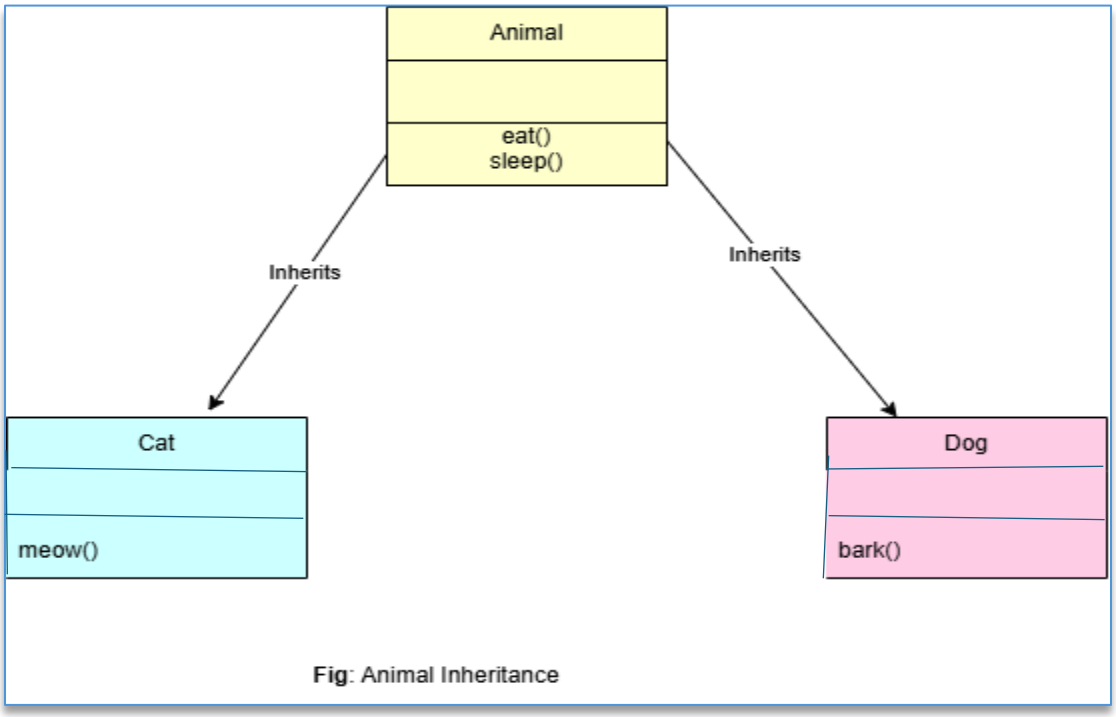


Fig: Animal Inheritance

Untitled1.cpp

```cpp
#include <iostream>
using namespace std;

class Animal {
public:
    void eat() {
        cout << "Animal is eating." << endl;
    }

    void sleep() {
        cout << "Animal is sleeping." << endl;
    }
};

class Cat : public Animal {
public:
    void meow() {
        cout << "Cat says: Meow!" << endl;
    }
};

class Dog : public Animal {
public:
    void bark() {
        cout << "Dog says: Woof!" << endl;
    }
};

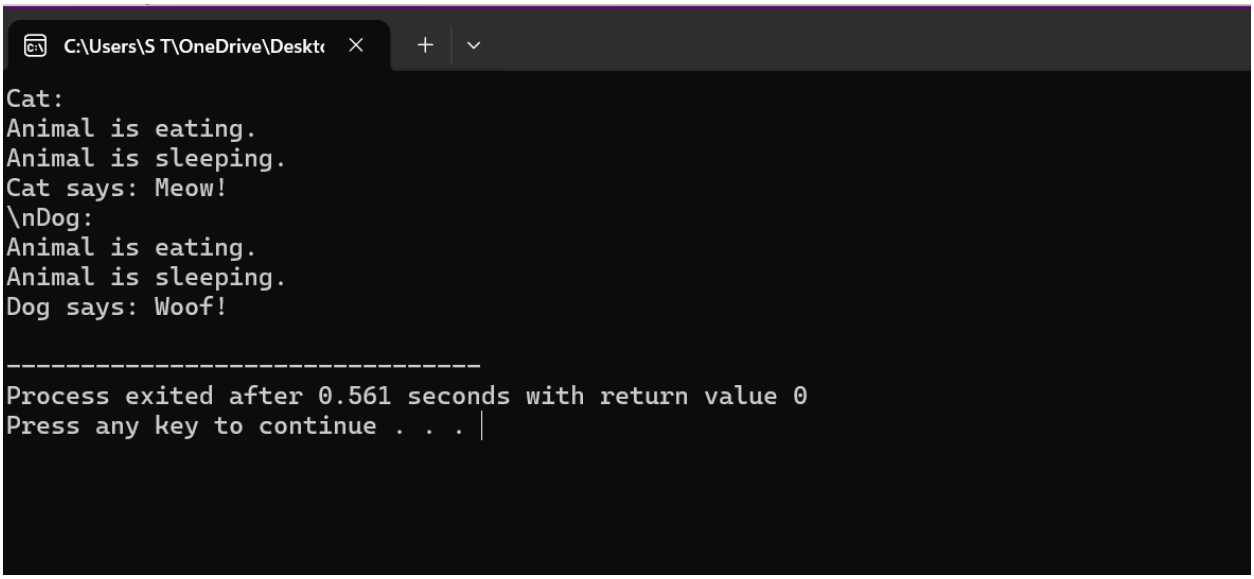
int main() {
    Cat myCat;
    Dog myDog;

    cout << "Cat:" << endl;
    myCat.eat();
    myCat.sleep();
    myCat.meow();

    cout << "\\nDog:" << endl;
    myDog.eat();
    myDog.sleep();
    myDog.bark();

    return 0;
}
```

Output



Q3. Shape, Triangle and Rectangle — Inheritance

Shape is the base class. Triangle and Rectangle inherit from Shape. The program takes the required measurements from the user and calculates the area of each shape. The Triangle area is (base × height) / 2, and the Rectangle area is length × width.

Untitled2.cpp

```cpp
#include <iostream>
using namespace std;

class Shape {
public:
    virtual double area() = 0; // pure virtual function
};

class Triangle : public Shape {
private:
    double base, height;

public:
    void input() {
        cout << "Enter triangle base: ";
        cin >> base;
        cout << "Enter triangle height: ";
        cin >> height;
    }

    double area() {
        return (base * height) / 2.0;
    }
};

```

Untitled2.cpp

```cpp

class Rectangle : public Shape {
private:
    double length, width;

public:
    void input() {
        cout << "Enter rectangle length: ";
        cin >> length;
        cout << "Enter rectangle width: ";
        cin >> width;
    }

    double area() {
        return length * width;
    }
};

int main()
    Triangle t;
    Rectangle r;

    cout << "TRIANGLE" << endl;
    t.input();
    cout << "Area of triangle = " << t.area() << endl;

    cout << "\\nRECTANGLE" << endl;
    r.input();
    cout << "Area of rectangle = " << r.area() << endl;

    return 0;
}
```

Output

```
C:\Users\S T\OneDrive\Deskt
TRIANGLE
Enter triangle base: 10
Enter triangle height: 5
Area of triangle = 25
\nRECTANGLE
Enter rectangle length: 8
Enter rectangle width: 4
Area of rectangle = 32

--------------------------------
Process exited after 34.6 seconds with return value 0
Press any key to continue . . .
```

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